

Sample Project Documentation

This document serves as the master reference for OptiCrop, consolidating key decisions, architecture, setup procedures, and team contributions into a single technical summary.

1. Project Summary

Project Name	OptiCrop Agricultural Crop Recommendation System
Track	AI-ML and GEN-AI Track
Academic Year	2026-2027
Problem Statement	Smallholder farmers lack access to data-driven crop advisory tools
Solution	ML-powered web app that recommends the optimal crop from 7 soil/climate inputs
Technology	Python 3.11, Flask 3.0, scikit-learn, Bootstrap 5, Jupyter Notebook
ML Model	Random Forest Classifier 99.09% test accuracy across 22 crop classes
Dataset	Kaggle Crop Recommendation Dataset 2,200 samples, 7 features, 22 labels
Deployment	Local Flask dev server (http://127.0.0.1:5000)

2. Input Parameters & Valid Ranges

Parameter	Symbol	Unit	Min	Max	Description
Nitrogen	N	mg/kg	0	140	Nitrogen content in soil essential for leaf growth
Phosphorus	P	mg/kg	5	145	Phosphorus content root development and flowering
Potassium	K	mg/kg	5	205	Potassium disease resistance and water regulation
Temperature	T	°C	8.0	44.0	Average ambient temperature during growing season
Humidity	H	%	14.0	100.0	Relative humidity affects transpiration and disease
pH Level	pH		3.5	9.5	Soil pH determines nutrient availability
Rainfall	R	mm	20.0	300.0	Average annual rainfall primary water source

3. Known Limitations & Future Work

Limitation	Impact	Mitigation / Roadmap
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Dataset limited to 22 crops and Indian conditions	Medium	Expand dataset; v2 includes 40+ crops
No real-time weather or NPK data integration	Medium	IoT sensor API v3.0 roadmap
English-only UI	Medium	Hindi/Marathi toggle v1.1 roadmap (Flask-Babel)
No user accounts or prediction history	Low	PostgreSQL + user auth v2.0 roadmap
Model not retrained with new data	Low	MLflow retraining pipeline v2.0 roadmap